

Yelp Business Analysis Report

1. Project Overview

This project analyzes Yelp business listing data using Python, MySQL, and Power BI. The objective is to identify customer engagement patterns, business performance, category trends, geographic distribution, and customer satisfaction levels. The insights generated from this analysis can help businesses improve visibility, customer experience, and decision-making.

2. Dataset Summary

Dataset Information

- Rows: 1,000,000
- Columns: 13
- Missing Values: 0
- Key Features:
 - Organization
 - Rating
 - NumberReview
 - Category
 - Country
 - State
 - City
 - Street
 - Building
 - Phone

Objective

The dataset was analyzed to identify top-performing businesses, highly rated categories, customer review trends, geographic business concentration, and business growth opportunities.

3. Data Cleaning and Preparation Using Python

The dataset was cleaned and prepared using Python and Pandas.

Steps Performed

- Loaded the dataset using Pandas.
 - Examined data using:
 - `df.head()`
 - `df.info()`
 - `df.describe()`
 - `df.isnull().sum()`
 - Removed unnecessary columns.
 - Standardized column names.
 - Filled missing values in:
 - Organization
 - State
 - City
 - Building
 - Street
 - Validated phone numbers and replaced invalid entries.
 - Checked duplicate records.
 - Exported the cleaned dataset.
 - Imported the cleaned dataset into MySQL for analysis.
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4. Data Analysis Using SQL

4.1 Which Are the Top 10 Most Reviewed Businesses?

Result

Buffalo Wild Wings, Panera Bread, Olive Garden Italian Restaurant, BJ's Restaurant & Brewhouse, Domino's Pizza, Gen Korean BBQ House, Wingstop, P.F. Chang's, Panda Express, and Popeyes Louisiana Kitchen.

Insight

Buffalo Wild Wings generated the highest review count, followed by Panera Bread and Olive Garden. These businesses demonstrate strong customer engagement and market presence. High review counts indicate strong brand recognition and a large customer base.

Business Impact

Businesses with higher customer engagement tend to attract more visibility and customer trust.

```
5 # Which are the top 10 most reviewed businesses?;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
organization	numberreviews		
▶ Buffalo Wild Wings	297080		
Panera Bread	234734		
Olive Garden Italian Restaurant	207039		
BJ's Restaurant & Brewhouse	171437		
Domino's Pizza	166864		
Gen Korean BBQ House	166459		
Wingstop	151568		
P.F. Chang's	144601		
Panda Express	133173		
Popeyes Louisiana Kitchen	131729		

4.2 Which Categories Have the Highest Number of Businesses?

Result

- Plumbing
- Delivery
- Air Conditioning & Heating

Insight

Plumbing dominates the dataset, indicating that home-service businesses form a significant portion of Yelp listings. Delivery services also represent a major segment due to increasing customer demand.

Business Impact

Service-oriented industries continue to maintain strong market demand and customer reach.

```
13 # Which categories have the highest number of businesses?
```

```
14
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
category	organizations		
▶ Plumbing	510305		
Delivery	348522		
Air Conditioning & Heating	141173		

4.3 Which Categories Have the Highest Average Ratings?

Result

- Delivery
- Air Conditioning & Heating
- Plumbing

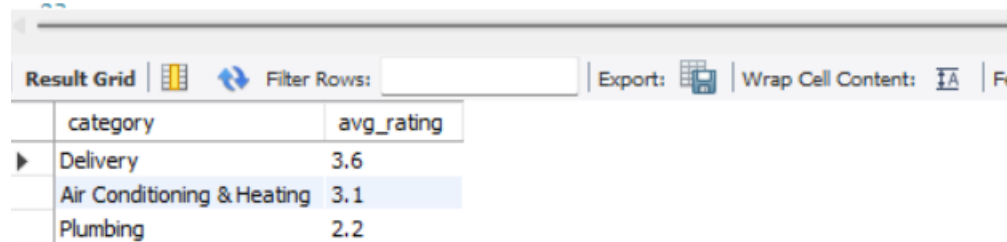
Insight

Delivery businesses received the highest average ratings, indicating stronger customer satisfaction and service quality. Plumbing businesses recorded comparatively lower ratings.

Business Impact

Higher ratings improve customer trust and increase the likelihood of customer acquisition.

```
21 # Which categories receive the highest average ratings?
22
23
```



category	avg_rating
Delivery	3.6
Air Conditioning & Heating	3.1
Plumbing	2.2

4.4 Which Categories Generate the Most Reviews?

Result

Delivery generated the highest review volume.

Insight

Customers interact more frequently with delivery-related businesses. This category shows the strongest customer engagement among all categories.

Business Impact

High engagement levels indicate strong demand and repeat customer interactions.

```
31 # Which categories generate the most customer reviews?
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	category	cus_review		
▶	Delivery	48744283		

4

.5 Which Cities Have the Largest Number of Businesses Listed?

Result

- Los Angeles
- Chicago
- New York

Insight

Los Angeles contains the highest concentration of businesses, followed by Chicago and New York. These cities represent major commercial centers.

Business Impact

Businesses operating in these cities benefit from larger customer markets and greater visibility.

```
39  
40 # Which cities have the largest number of businesses listed?  
41
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	city	org		
▶	Los Angeles	13898		
	Chicago	11733		
	New York	9365		

4.6 Which States Have the Highest Concentration of Businesses?

Result

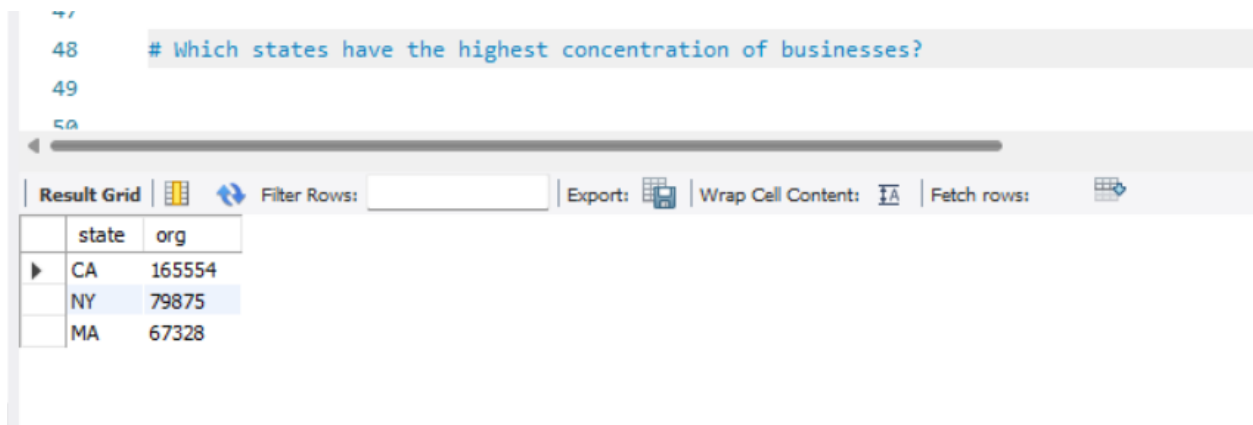
- California
- New York
- Massachusetts

Insight

California contains the highest concentration of businesses in the dataset. The state provides significant business opportunities and customer reach.

Business Impact

Large business ecosystems create opportunities for growth and competition.



The screenshot shows a data analysis interface. At the top, a query is entered: `# Which states have the highest concentration of businesses?`. Below the query, a horizontal scrollbar is visible. Underneath the scrollbar is a toolbar with the following options: 'Result Grid' (selected), 'Filter Rows:' (with a search input), 'Export:' (with a download icon), 'Wrap Cell Content:' (with a text icon), and 'Fetch rows:' (with a refresh icon). Below the toolbar is a table with the following data:

	state	org
▶	CA	165554
	NY	79875
	MA	67328

4.7 Which Cities Have the Highest Average Ratings?

Insight

Several cities achieved perfect average ratings, indicating exceptional customer satisfaction. However, some locations may contain fewer businesses and review counts.

Business Impact

Highly rated cities can serve as benchmarks for customer experience and service quality.

```

57 # Which cities have the highest average business ratings?
58
59

```

city	avg_rating
Shannon	5
Boyle Heights	5
Los Nietos	5
Weatminster	5
Silverlake	5
Avocado Heights	5
Rossmoor	5
Woodland Hls	5

4.8 Is There a Relationship Between Ratings and Number of Reviews?

Result

Excellent-rated businesses generated the highest review volume.

Insight

Customers are more likely to engage with highly rated businesses. Better ratings generally lead to increased visibility and customer interaction.

Business Impact

Maintaining high ratings can directly influence customer engagement and business growth.

```

75
76 # Is there a relationship between ratings and number of reviews?
77

```

review_rating	total_review
excellent review	39218305
good review	14530445
poor review	2936867

4.9 Which Businesses Have High Ratings but Relatively Few Reviews (Hidden Gems)?

Result

Several businesses achieved perfect ratings while receiving very few reviews.

Insight

These businesses offer excellent service quality but have limited visibility among customers.

Business Impact

Marketing efforts and review-generation strategies can help these businesses grow.

```
90 # Which businesses have high ratings but relatively few reviews (hidden gems)?
91
92
```

organization	ratings	total_review
Air Care Cooling and Heating	5	1
Ocean Air Condo	5	1
Vision Appliance's & Air conditioning	5	1
Royalty Mechanical Heating and Air conditioning	5	1
All County Mechanical	5	1
Bob's Heating and Air Conditioning	5	1
Chas F Ober & Son	5	1
Quick Repair Heating & Air	5	1

Result 14

4.10 Which Businesses Have Many Reviews but Low Ratings?

Result

Examples include:

- Buffalo Wild Wings
- Panera Bread
- Olive Garden
- Domino's Pizza

Insight

These businesses attract significant customer traffic but struggle with customer satisfaction.

Business Impact

Improving service quality can help improve ratings while maintaining review volume.

```
100
101 # Which businesses have many reviews but low ratings (customer satisfaction issues)?
102
103
```

organization	ratings	avg_review
Buffalo Wild Wings	1	297080
Panera Bread	1	234734
Olive Garden Italian Restaurant	2	207039
Domino's Pizza	1	166864
Wingstop	1	151568
Panda Express	1	133173
Popeyes Louisiana Kitchen	1	131729
Jersey Mike's Subs	1	107019

4.11 What Percentage of Businesses Have Ratings Above 4.0?

Result

29.21%

Insight

Only around 29% of businesses maintain ratings above 4.0. High-performing businesses represent a relatively small segment of the market.

Business Impact

Maintaining ratings above 4.0 provides a strong competitive advantage.

```
111
112 # What percentage of businesses have ratings above 4.0?
113
```

	count(organization)	percentage_of_bussiness
▶	292134	29.2134

4.12 Which City Contributes the Highest Share of Total Reviews?

Result

- Los Angeles
- New York
- San Francisco

Insight

Los Angeles contributes the largest share of reviews, indicating extremely high customer activity and engagement.

Business Impact

Businesses in highly active cities have greater opportunities to build brand awareness.

```
110
119
120 # Which city contributes the highest share of total reviews?
121
```

	city	total_review
▶	Los Angeles	2515438
	New York	1597165
	San Francisco	1049053

4.13 Which Category Dominates in Each State?

Insight

The dominant category differs across states, reflecting regional customer needs and market demand.

Business Impact

Businesses should adapt strategies based on regional market characteristics.

```
129
130 # Which category dominates in each state?
131
```

	state	category	total_businesses
▶	10	Air Conditioning & Heating	1
	48	Delivery	2
	ABD	Plumbing	2
	ABE	Plumbing	5
	AK	Delivery	312
	AL	Plumbing	7981
	Apple	Plumbing	4
	AR	Plumbing	5066

4.14 Which Business Category Shows the Best Combination of High Ratings and High Review Volume?

Result

Delivery

Insight

Delivery demonstrates the strongest balance between customer satisfaction and customer engagement.

Business Impact

Delivery businesses provide a strong example of successful customer-focused operations.

149

150 # Which business category shows the best combination of high ratings and high review volume?

151

Result Grid





Filter Rows:

Export:



Wrap Cell Content:



	category	avg_rating	total_review
▶	Delivery	3.55	48744283
	Air Conditioning & Heating	3.13	2654007
	Plumbing	2.16	5287327

5. Dashboard in Power BI

KPI Overview

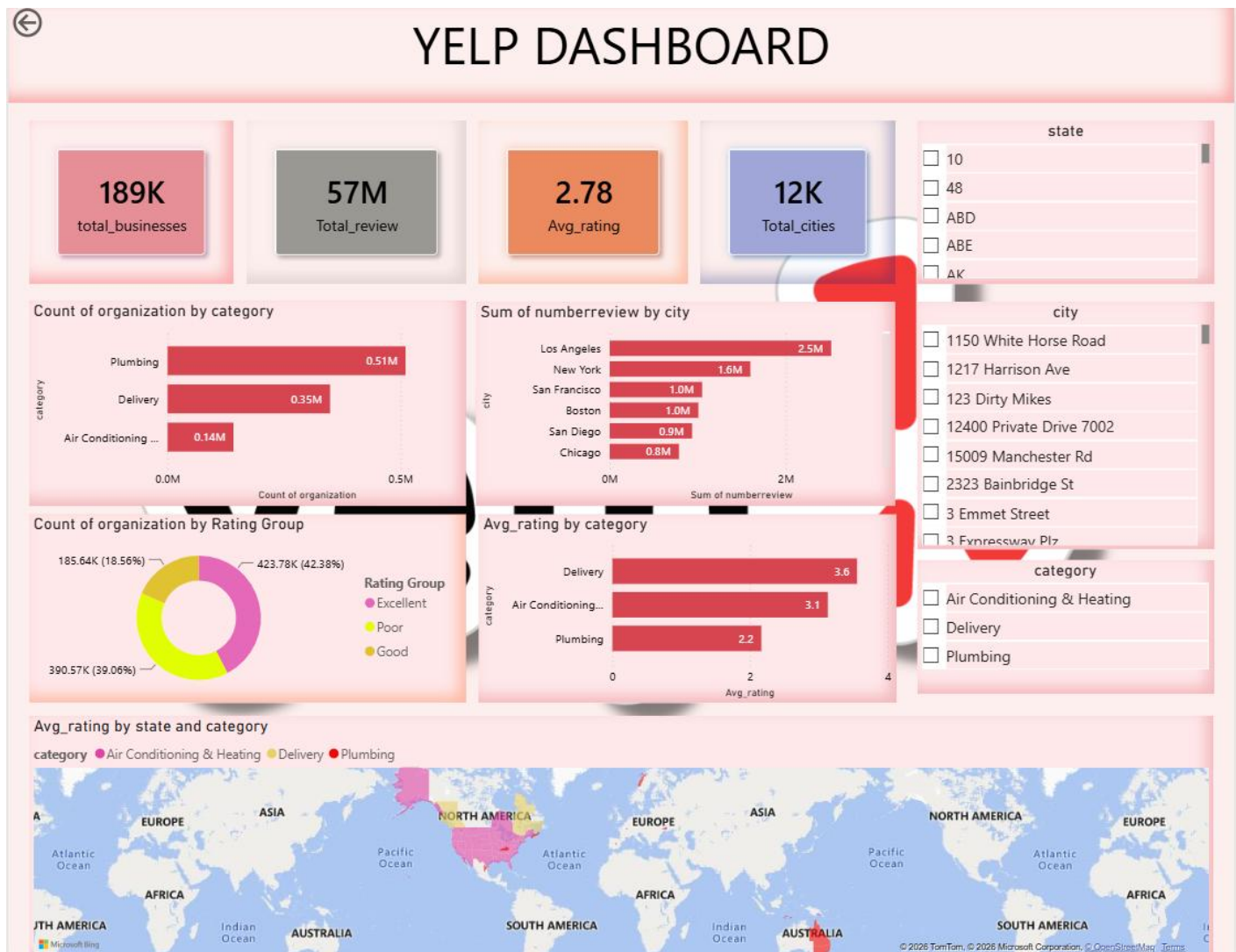
- Total Businesses: 189K
- Total Reviews: 57M
- Average Rating: 2.78
- Total Cities: 12K

Dashboard Insights

- Plumbing contains the highest number of businesses.
- Los Angeles generates the highest review volume.
- Delivery has the highest average rating.
- Excellent-rated businesses account for the majority of reviews.
- California shows the highest concentration of businesses.

Dashboard Benefits

The dashboard allows users to explore business performance, customer engagement, ratings, categories, and geographic trends interactively.



6. Business Recommendations

Promote Hidden Gems

Increase visibility of highly rated businesses with low review counts.

Improve Customer Satisfaction

Focus on businesses with low ratings but high review volume.

Expand Delivery Strategies

Adopt successful delivery-service practices across other categories.

Encourage Customer Reviews

Actively collect customer reviews to improve visibility and trust.

Focus on High-Engagement Cities

Target marketing efforts in Los Angeles, New York, and San Francisco.

7. Conclusion

The Yelp Business Analysis project successfully identified trends in customer engagement, ratings, category performance, and geographic distribution. Python was used for data preparation, MySQL for business analysis, and Power BI for visualization. The findings reveal that Delivery is the strongest-performing category, Los Angeles is the most active market, and customer ratings significantly influence engagement. These insights can help businesses improve customer satisfaction, increase visibility, and make data-driven decisions.